BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q.PAPER – II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8195

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If $f(x) = 3 - \sqrt{x}$ then $f'(1)$ is equal to :

(A) -1/2 (B) 0

(C) 1/2 (D) 1

Q2. $4\int_0^{\pi/4} \sin 2x \, dx =$:

(A) $4 - 2\sqrt{2}$ (B) $\sqrt{3}/2$

(C) 1/2 (D) $\sqrt{3}$

Q3. $\lim_{x\to 0} \sin ax / \sin bx =$:

(A) -a/b (B) -b/a

(C) a/b (D) b/a

Q4. $\int \ln x \, dx =$:

(A) $1/x + c$ (B) $x \ln x + c$

(C) $(\ln x)^2/2 + c$ (D) $x(\ln x - 1) + c$

Q5. Let $f(x) = \sqrt{(1 - x^2)}$ in R then domain of f is :

(A) Real numbers (B) $|x| \leq 1$

(C) Negative real numbers (D) Integers

Q6. If $\int x e^{x^2} dx = k e^{x^2}$ then k = :

(A) 1/2 (B) 1/3

(C) x/3 (D) x/2

Q7. If f(x) has second derivative at c such that f'(c) = 0 and f''(c) < 0 then c is point of :

(A) Maxima (B) Minima

(C) Point of inflection (D) Origin

Q8. If y = cot x, then dy/dx is given by :

(A) $\text{cosec}^2 x$ (B) $-\text{cosec}^2 x$

(C) tan x (D) -cosec x cot x

Q9. $\int 1/(x^2 + a^2) dx$ = :

(A) $\tan^{-1}(x/a) + c$ (B) $(1/a) \tan^{-1}(x/a) + c$

(C) $(a/x) \tan^{-1}(x/a) + c$ (D) $(1/a) \tan^{-1}(a/x) + c$

Q10. For $y = \log_e 5x$, dy/dx = :

(A) 1/x (B) 5

(C) 1/(5x) (D) 1

Q11. The straight line y = mx + c is tangent to the parabola $y^2 = 4ax$ if :

(A) c = a/m (B) c = m/a

(C) $c = a^2/m^2$ (D) c = am

Q12. y-coordinate of any point on x-axis is :

(A) 0 (B) x

(C) 1 (D) y

Q13. The volume of parallelepiped determined by u = i + 2j - k , v = i - 2j + 3k , w = i - 7j - 4k is :

(A) 48 (B) 50

(C) 52 (D) 55

Q14. The distance between the centres of the circles $x^2 + y^2 + 2x + 2y + 1 = 0$ and $x^2 + y^2 - 4x - 6y - 3 = 0$ is :

(A) 1 (B) 4

(C) 5 (D) 15

Q15. If a + b + c = 0 then which one is correct :

(A) $a \times b \times c = 0$ (B) $a \times b = b \times c = c \times a$

(C) a.b = b.c = c.a (D) a = b = c

Q16. The x-intercept of the line $2x + 3y - 1 = 0$ is :

(A) 2 (B) 3

(C) 1/3 (D) 1/2

Q17. The graph of $2x - 3y \leq 6$ is :

(A) On the origin side (B) Not on the origin side

(C) Not decided (D) Through the origin

Q18. The area of the triangle having a and b as its two sides is given by :

(A) $|a.b|$ (B) $(1/2)|a.b|$

(C) $|a \times b|$ (D) $(1/2)|a \times b|$

Q19. Homogeneous equation of second degree $ax^2 + 2hxy + by^2 = 0$ where a, b, h are not all zero, represents two imaginary lines if :

(A) $h^2 = ab$ (B) $h^2 > ab$

(C) $h^2 < ab$ (D) $h = ab$

Q20. The eccentricity of the ellipse $x^2/64 + y^2/28 = 1$ is :

(A) 3/4 (B) 4/3

(C) $\sqrt{(3/4)}$ (D) $\sqrt{(4/3)}$

PAPER – II (Essay Type)

Time Allowed: 2.30 hours | **Maximum Marks:** 80

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Prove that $\cosh^2 x - \sinh^2 x = 1$

(ii) If $f(x) = \sqrt{(x + 4)}$ then find $f(x - 1)$

(iii) Evaluate $\text{Lim}_{x \to 3} (x - 3)/(\sqrt{x} - \sqrt{3})$

(iv) Evaluate $\text{Lim}_{x \to 0} (1 - \cos 2x)/x^2$

(v) Differentiate $y = (x^2 + 5)(x^3 + 7)$ with respect to x.

(vi) Differentiate $(x^2 + 1)/(x^2 - 3)$ with respect to x.

(vii) Find derivative of $(x^3 + 1)^9$ with respect to x.

(viii) Differentiate $\cos\sqrt{(x + \sqrt{\sin x})}$ with respect to the variable involved.

(ix) dy/dx = ? If $y = e^{x^2} + 1$

(x) Find Maclaurin Series for sin x

(xi) Determine the interval in which $f(x) = 4 - x^2$, $x \in (-2, 2)$ is increasing or decreasing.

(xii) Find f'(x) if f(x) = √(ln(e^{2x} + e^{-2x}))

Q3. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Using differential to find dy/dx if xy + x = 4

(ii) Evaluate ∫(a - 2x)^{3/2} dx

(iii) Evaluate ∫ sec x dx

(iv) Evaluate ∫ x ln x dx

(v) Evaluate $\int_1^2 x/(x^2 + 2)\, dx$

(vi) Find the area bounded by cos function from $x = -\pi/2$ to $x = \pi/2$

(vii) Solve the differential equation $dy/dx = y/x^2$

(viii) Find h such that A(-1, h), B(3, 2) and C(7, 3) are collinear.

(ix) The coordinates of a point P are (3, 2). The axes are translated through the point O'(1, 3). Find the coordinates of P referred to new axes.

(x) Find k so that the line joining A(7, 3); B(k, -6) and the line joining C(-4, 5); D(-6, 4) are parallel.

(xi) Find the point of intersection of the lines x - 2y + 1 = 0 and 2x - y + 2 = 0

(xii) Find measure of the angle between the lines represented by $9x^2 + 24xy + 16y^2 = 0$

Q4. (Marks 18) Write short answers to any NINE (9) questions :

(i) Graph the solution set of inequality $3x - 2y \geq 6$

(ii) Define feasible region.

(iii) Find the equation of circle whose ends of diameter are (-3, 2) and (5, -6)

(iv) Find the position of the point (5, 6) w.r.t the circle $2x^2 + 2y^2 + 12x - 8y + 1 = 0$

(v) Find the focus and vertex of parabola $y^2 = -8(x - 3)$

(vi) Find the eccentricity of ellipse $x^2 + 4y^2 = 16$

(vii) Find the centre and eccentricity of the conic $y^2/4 - x^2 = 1$

(viii) Identify the conic represented by $4x^2 - 4xy + y^2 - 6 = 0$

(ix) Find the work done by a constant force F = 2i + 4j, if its point of application to a body moves it from A(1, 1) to B(4, 6)

(x) Find the value of 'α' such that αi + j , i + j + 3k and 2i + j - 2k are coplanar.

(xi) If u = 2i - j + k and v = 4i + 2j - k find u × v

(xii) Find a vector whose magnitude is 4 and is parallel to 2i - 3j + 6k

(xiii) If A(1, -1), B(2, 0), C(-1, 3) and D(-2, 2) are given points, find the sum of the vectors AB and CD

> Note: SECTION – II

> Note: Note : Attempt any THREE questions.

Q5.

(a) [5] Find m and n , so that given function f is continuous at x = 3 ; f(x) = { mx if x < 3 ; n if x = 3 ; -2x + 9 if x > 3 }

(b) [5] Prove that y dy/dx + x = 0 if x = $(1 - t^2)/(1 + t^2)$, y = $2t/(1 + t^2)$

Q6.

(a) [5] If y = e^{-ax}, then show that $d^3y/dx^3 + a^3y = 0$

(b) [5] Evaluate the indefinite integral $\int\sqrt{(x^2 - a^2)}\, dx$

Q7.

(a) [5] Solve the differential equation 2e^x tan y dx + (1 - e^x) $\sec^2$ y dy = 0

(b) [5] Maximize f(x, y) = x + 3y subject to the constraints 2x + 5y ≤ 30 ; 5x + 4y ≤ 20 , x ≥ 0 , y ≥ 0

Q8.

(a) [5] Find equations of the tangents to the circle $x^2 + y^2 = 2$ perpendicular to the line 3x + 2y = 6

(b) [5] Using vectors, prove that sin(α + β) = sin α cos β + cos α sin β

Q9.

(a) [5] Find centre, foci, eccentricity, vertices and equation of directrices of $(y + 2)^2/9 - (x - 2)^2/16 = 1$

(b) [5] Find the equations of altitudes of the triangle whose vertices are A(-3, 2), B(5, 4), C(3, -8)

GROUP – II

Q.PAPER – II (Objective Type)

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 8192

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If $f(x) = 1/x^2$ then which of the following is equal to fof(x) :

(A) x^4 (B) x^2

(C) 1 (D) $1/x^4$

Q2. What is the value of Lim_{x→0} (x sin x) :

(A) α (B) -1

(C) 1 (D) 0

Q3. What is the value of $\sqrt{(1 - x^2)}$ d/dx ($\sin^{-1}x + \cos^{-1}x$) :

(A) $\sqrt{(1 - x^2)}$ (B) 0

(C) 2 (D) 1/x

Q4. d/dx ($\sinh^{-1}x$) = :

(A) $1/\sqrt{(1 - x^2)}$ (B) $-1/\sqrt{(1 - x^2)}$

(C) $1/\sqrt{(1 + x^2)}$ (D) $-1/\sqrt{(1 + x^2)}$

Q5. Derivative of x^3 w.r.t x^3 is :

(A) 0 (B) 1

(C) x^3 (D) $3x^2$

Q6. If $f(x) = a^x$ then f'(x) = :

(A) $a^x \ln a$ (B) $a^x \ln x$

(C) $a^x(\ln a)^2$ (D) $(a^x)^2 \ln a$

Q7. $\int x^{-1} dx$:

(A) 0 (B) $-x^{-2} + c$

(C) ∝ (D) $\ln x + c$

Q8. $\int_0^1 1/\sqrt{(1 - x^2)} dx$ = :

(A) π/6 (B) π/4

(C) π/3 (D) π/2

Q9. $\int \tan x dx$ = :

(A) ln cot x + c (B) ln sec x + c

(C) ln sin x + c (D) ln cosec x + c

Q10. $\int_0^\pi \sin x dx$ = :

(A) 0 (B) 1

(C) 2 (D) π

Q11. A linear equation in two variables represents :

(A) Circle
(B) Ellipse
(C) Hyperbola
(D) Straight line

Q12. Intercept form of equation of line is :

(A) $x/a + y/b = 1$
(B) $x/a + y/b = 0$
(C) $x/a - y/b = 1$
(D) $x/a - y/b = 0$

Q13. Distance of point (cos 3x, sin 3x) from origin is :

(A) 3
(B) 6
(C) 9
(D) 1

Q14. (0, 0) is one of the solution of inequality :

(A) $3x + 5y > 4$
(B) $2x + 3y < 4$
(C) $x + 3y > 5$
(D) $2x + 3y > 5$

Q15. Equation of circle with centre (3, 0) and radius $\sqrt{9}$ is :

(A) $x^2 + y^2 - 6x = 0$
(B) $x^2 - 6x = 9$
(C) $x^2 + y^2 = 9$
(D) $9x^2 + y^2 = 9$

Q16. Equation of directrix of parabola $y^2 = -12x$ is :

(A) $x = -3$
(B) $x = 3$
(C) $y = 3$
(D) $y = -3$

Q17. Co-vertices of ellipse $x^2/b^2 + y^2/a^2 = 1$; $a > b$ are :

(A) $(\pm a, 0)$
(B) $(0, \pm a)$
(C) $(0, \pm b)$
(D) $(\pm b, 0)$

Q18. Which of the following vectors is equal to the vector $i.j \times k$:

(A) 0
(B) 1
(C) -1
(D) i

Q19. For what value of P [2 P 5] is perpendicular to [3 1 P] :

(A) 2/3
(B) -1
(C) 1
(D) $\sqrt{5}$

Q20. If a and b are parallel vectors then $a \times b =$:

(A) 0
(B) 1
(C) -1
(D) 2

PAPER – II (Essay Type)

Time Allowed: 2.30 hours | **Maximum Marks:** 80

Note: SECTION – I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Given that f(x) = cos x find (f(a + h) - f(a))/h and simplify.

(ii) If $f(x) = (-x + 9)^3$, find $f^{-1}(x)$

(iii) By rationalizing, find Lim_{x→0} (√(x + a) - √a)/x

(iv) Write down the domain and range of f(x) = 2x - 5

(v) Calculate derivative of f(x) = x^{2/3} at x = 8

(vi) Find derivative of (1 + x)/(1 - x) w.r.t. x

(vii) If $y = x^4 + 2x^2 + 2$, find dy/dx

(viii) Find dy/dx of implicit function $x^2 - 4xy - 5y = 0$

(ix) Apply chain rule to find dy/du if $y = x^2 + 1/x^2$ and u = x - 1/x

(x) Differentiate $\sin^2 x$ w.r.t $\cos^4 x$

(xi) Find f'(x) if $f(x) = x^3 e^{1/x}$

(xii) Find y_2 if $y = x^2.e^{-x}$

Q3. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Using differential to find dx/dy of $x^4 + y^2 = xy^2$

(ii) Evaluate ∫(2x + 3)^{1/2} dx

(iii) Evaluate ∫x√(x - a) dx

(iv) Evaluate $\int (\ln x)^2 dx$

(v) Evaluate $\int_1^2 (x + 1/x)^{1/2} (1 - 1/x^2) dx$

(vi) Find the area bounded by cos function from x = -π/2 to x = π/2

(vii) Solve dy/dx + 2xy/(2y + 1) = x

(viii) Find the mid-points of the line joining the two points A(-8, 3), B(2, -1).

(ix) Find h such that the points A(-1, h), B(3, 2) and C(7, 3) are collinear.

(x) In the triangle A(8, 6), B(-4, 2), C(-2, -6), find the slope of altitude of triangle.

(xi) Using slopes, show that the triangle with vertices A(6, 1), B(2, 7), C(-6, -7) is a right triangle.

(xii) Find the point of intersection of the lines $x + 4y - 12 = 0$ and $x - 3y + 3 = 0$

Q4. (Marks 18) Write short answers to any NINE (9) questions :

(i) Define feasible region.

(ii) Graph the solution set of $5x - 4y \leq 20$

(iii) Write the standard and general equation of circle.

(iv) Find centre and radius of $5x^2 + 5y^2 + 24x + 36y + 10 = 0$

(v) Check the position of the point (5, 6) with respect to the circle $x^2 + y^2 = 81$

(vi) Find the length of the tangent drawn from the point (-5, 4) to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$

(vii) Find foci and eccentricity of ellipse $x^2 + 4y^2 = 16$

(viii) Find the points of intersection of $x^2 + y^2 = 8$ and $x^2 - y^2 = 1$

(ix) If $u = 2i - 7j$, $v = i - 6j$ and $w = -i + j$, find $(1/2)u + (1/2)v + (1/2)w$

(x) Find a vector whose magnitude is 4 and is parallel to $2i - 3j + 6k$

(xi) Find α so that the vector u and v are perpendicular; $u = \alpha i + 2\alpha j - k$ and $v = i + \alpha j + 3k$

(xii) Find the area of parallelogram whose vertices are A(1, 2, -1); B(4, 2, -3); C(6, -5, 2); D(9, -5, 0)

(xiii) Prove that $u.(v \times w) + v.(w \times u) + w.(u \times v) = 3u.(v \times w)$

> Note: SECTION – II

> Note: Note : Attempt any THREE questions.

Q5.

(a) [5] Evaluate Lim_{x→0} (sec x - cos x)/x

(b) [5] Find the derivative w.r.t. x $\sin\sqrt{((1 + 2x)/(1 + x))}$

Q6.

(a) [5] If $y = (\cos^{-1}x)^2$, prove that $(1 - x^2)y_2 - xy_1 - 2 = 0$

(b) [5] Evaluate $\int(2x/(1 - \sin x))\, dx$

Q7.

(a) [5] Find the area between the x-axis and the curve $y = \sqrt{(2ax - x^2)}$ when $a > 0$

(b) [5] Maximize $f(x, y) = 2x + 5y$ subject to the constraints $2y - x \leq 8$; $x - y \leq 4$, $x \geq 0$, $y \geq 0$

Q8.

(a) [5] Find equation of the circle passing through the points A(3, -1), B(0, 1) and having centre at $4x - 3y - 3 = 0$

(b) [5] Use vectors to prove that $\sin(\alpha - \beta) = \sin\alpha \cos\beta - \cos\alpha \sin\beta$

Q9.

(a) [5] Mid-points of sides of triangle are (1, -1), (-4, -3) and (-1, 1). Find coordinates of vertices of triangle.

(b) [5] Show that equation of parabola with focus at $(a\cos\alpha, a\sin\alpha)$ and directrix $x\cos\alpha + y\sin\alpha + a = 0$ is $(x\sin\alpha - y\cos\alpha)^2 = 4a(x\cos\alpha + y\sin\alpha)$